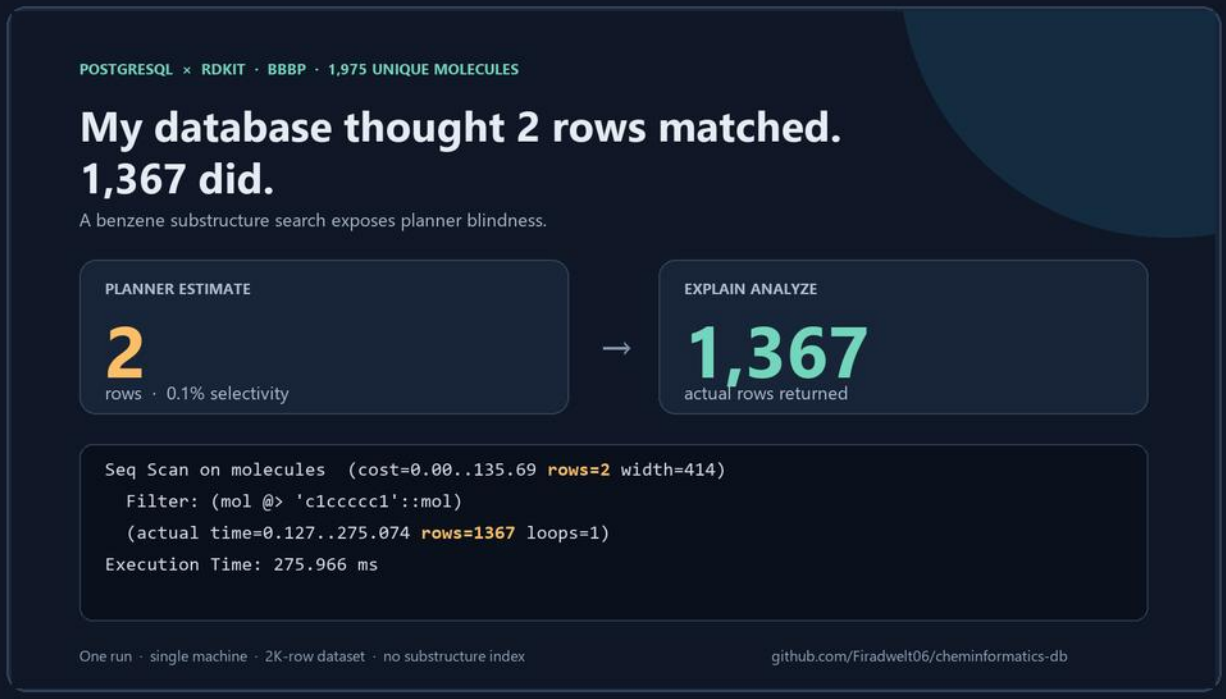
POSTGRESQL × RDKIT · BBBP

Estimated vs. actual.

A benzene-ring substructure search.



2 estimated rows → 1,367 actual

The plan screenshot is the evidence; the mismatch is the hook.

Single machine · small dataset · no substructure index

Store the parsed structure?

PARSE SMILES PER ROW

0.30–0.35

ms per row

STORED RDKit mol

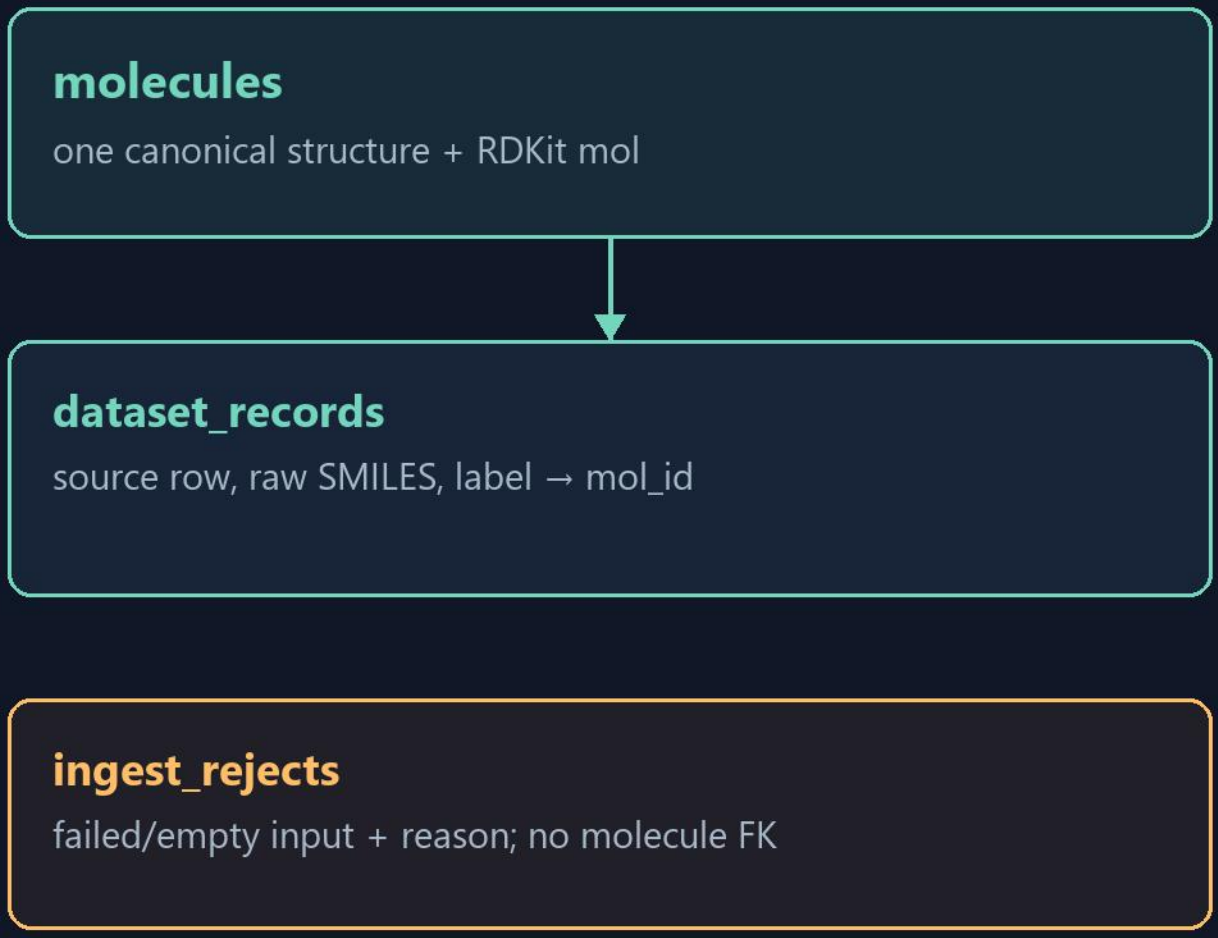
~0.06

ms per row · about 5–6× faster

But the mol value uses ~357 B vs 52 B for SMILES: about 6.9× more per value.

Per-value storage; table/index overhead excluded.

Keep chemistry and labels in the right places.



A real FK links dataset_records.mol_id → molecules.mol_id.

Same canonical molecule. Two different labels.

mol_id 961 · source row 98 · label 0

```
CC(=O)Oc1ccccc1C(O)=O
```

---

mol_id 961 · source row 996 · label 1

```
C1=CC=CC(=C1C(O)=O)OC(C)=O
```

Different SMILES traversals of aspirin.

Hypothesis: label/source noise. The cause needs source-record verification.

10 conflicting groups observed; cause not established.

WHAT COMES NEXT

2K rows is a baseline.
Week 2 is the real test.

01 Build and measure substructure indexes

02 Test parallel workers

03 Scale the benchmark to 10M rows

Current caveats: one machine, ±10–15% timing noise, indexes not built yet.

github.com/Firadwelt06/cheminformatics-db

Follow along: indexes · parallel workers · 10M rows